

Risk-Aware Investment in a Forest-Residue-to-SAF Supply Chain under Price Uncertainty

A Welfare-Theoretic Stochastic-Programming Analysis

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Forthcoming, *Energy Economics*

Road map

- I. **Motivation** — SAF Grand Challenge, the under-the-uncertainty siting problem
- II. **Welfare framework** — planner, make-vs-buy, mean-CVaR
- III. **Closed-form benchmark** — toy 2-state model, $\sigma^* = \lambda \pi_H (a_H - a_L)$
- IV. **Empirical SP-MILP** — 248 depots / 34 refineries, EVPI, CVaR tail
- V. **Policy bridge & conclusion** — σ_{KDE}^* vs. 45Z

Target audience: welfare economists, energy/biofuel policy, supply-chain SP researchers.

Act I — Motivation

The SAF Grand Challenge as an economic problem

U.S. policy target: 3 B gallons of sustainable aviation fuel by 2030; 35 B gallons by 2050.

- Current actuals: $\sim 0.5\text{--}0.7$ B gal/yr (EIA STEO 2025)
- Mature FT capacity, ESG offtake, 45Z credit *persistence* required to close the gap
- Decision horizon: 2026–2035 build-out window

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The empirical setting: southeastern U.S. forest residue.

- 12 SE-U.S. states, 14,040 hexes (50 mi^2), aggregated to 1,374 counties
- Hardwood-residue pool: ~ 25.8 M dt/yr (within 2023 Billion-Ton Report ceiling)
- Pre-processing depots + biorefineries to be sited *here-and-now*

Why is this not just an engineering siting exercise?

SAF wholesale price is volatile.

- EIA wholesale Gulf-Coast jet-A spot: \$3.81–\$9.12/gal (2014–2024)
- Empirical mean $\sim$ \$6.00/gal, KDE-fitted support [\$3.81, \$9.12]
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Existing literatures, each with a gap:

Bioenergy SP-MILPs (Eksioglu 2009; Marvin 2012; Quddus 2018) — rich spatial detail, but no welfare framing

Risk-functional SP (Rockafellar–Uryasev 2000; Conejo 2010) — right risk machinery, but never joined to landscape-scale siting

Prices vs. quantities (Weitzman 1974) — the right *welfare* object, but risk-neutral and one-period

This paper's research question

Question

Under SAF wholesale-price uncertainty, what is the welfare-frame per-gallon subsidy — and the EVPI, and the CVaR-tail loss — that a benevolent SE-U.S. planner faces when siting a residue-to-SAF supply chain?

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Three contributions:

1. **Conceptual.** Recast a familiar engineering MILP as a social-planner welfare program with explicit make-vs-buy slack at \$3.50/gal.
2. **Methodological.** Closed-form 2-state benchmark $\sigma^* = \lambda\pi_H(a_H - a_L) + \text{Madansky EVPI corollary } \text{Var}(A)/(2K)$.
3. **Empirical.** 700,000-variable SP solved at FIPS-5 county resolution; EVPI reported as a *candidate-based estimate with disclosed bias channels*.

Headline numbers (preview)

Quantity	Value	Reading
σ_{KDE}^*	\$0.52/gal	welfare-frame risk-premium subsidy
EVPI (\$/yr, single-year-equiv.)	$\approx$ \$255M	19.8% of $\mathbb{E}[\text{PVC}]$, candidate-based estimate
$\mathbb{E}[L]$ vs $\text{CVaR}_{0.95}[L]$	\$299M \$921M	vs. $3\times$ tail/mean, certified eval.
Siting	248 / 34	depots / refineries across 10 states
$\text{RP} \subset \text{WS-union}$	100%	every RP depot appears in some WS

All NPV at $\rho = 10\%$ over 2026–2035; 2026 USD throughout.

Act II — Welfare Framework

The planner's primitives

Sets.

- $\mathcal{H}$: residue-source counties — $|\mathcal{H}| = 1,374$
- $\mathcal{D}$: candidate depot counties — $|\mathcal{D}| = 528$
- $\mathcal{R}$: candidate biorefinery counties — $|\mathcal{R}| = 528$
- $\mathcal{S}$: SAF-price scenarios — $|\mathcal{S}| = 20$ (KDE bins)

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Timing (two-stage SP).

1. Choose binaries $y \in \{0, 1\}^{|\mathcal{D}|+|\mathcal{R}|}$ *here and now*
2. Nature draws $\omega \in \mathcal{S}$: $p_{\text{SAF}}(\omega)$ and ash perturbation $\sigma(\omega)$
3. Choose flows $\mathbf{x}(\omega), \mathbf{z}(\omega) \geq 0$ and slack-import $s(\omega) \geq 0$

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Risk. Mean-CVaR functional with weight $\lambda \in [0, 1]$, confidence $\alpha \in (0, 1)$.

Welfare components

For output Q in scenario ω at price $p_{\text{SAF}}(\omega)$:

$$W(\mathbf{x}, \mathbf{z}, y; \omega) = \underbrace{\text{CS}(Q; \omega)}_{\text{consumer surplus}} + \underbrace{\text{PS}(\mathbf{x}, \mathbf{z}, y; \omega)}_{\text{producer surplus}} - \underbrace{p_{\text{SAF}}^f \cdot s(\omega)}_{\text{make-vs-buy slack}}$$

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Make-vs-buy is load-bearing.

- Imported wholesale jet at $p_{\text{SAF}}^f = \$3.50/\text{gal}$ (long-run anchor; STEO 2014–2024)
- Without it: unmet demand is costless $\Rightarrow$ trivializes the build-vs-no-build decision
- **Welfare quantity**, not a producer cost — this is the gap relative to Eksioglu (2009) etc.

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Carbon externality held at zero in this paper ($p_C = 0$): isolates the *price-uncertainty channel*. The 45Z + carbon-tax extension is the companion paper.

Stochastic welfare-maximization problem

$$\max_{\substack{y \in \{0,1\}^{|\mathcal{D}|+|\mathcal{R}|} \\ \mathbf{x}, \mathbf{z}, s \geq 0}} \mathbb{E}_{\omega} [\text{CS}(\mathbf{z}; \omega) + \text{PS}(\mathbf{x}, \mathbf{z}, y; \omega) - p_{\text{SAF}}^f s(\omega)]$$

subject to mass balance, ash-tier bounds, capacity, and market clearing.

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Risk-averse variant. Replace $\mathbb{E}[\cdot]$ with

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$\Rightarrow$ MILP gets $|\mathcal{S}| + 1$ extra variables $(\eta, \{\zeta_{\omega}\})$ via Rockafellar–Uryasev (2000).

We hold these exogenous:

1. CGE stumpage-price endogeneity (Hertel et al. 2010)
2. LCFS CI-segmentation rents (Holland et al. 2009): $\sim \$0.20\text{--}0.35/\text{gal}$
3. RFS RIN pass-through (Lade & Bushnell 2019): 50–70% short-run
4. Indirect land-use change (residue feedstock partially shielded)

Scope — partial equilibrium and four GE channels

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Implication. The bound is *two-sided* (paper §6.1): financing frictions and hurdle wedges push σ^* up; offtake coverage, cost–price co-movement, and partial hedging push it down.

$\sigma_{\text{KDE}}^* = \$0.52/\text{gal}$: lower bound in the λ -channels, upper bound in the coverage/hedging channels

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A CGE-consistent welfare evaluation is the natural next step — flagged in conclusion.

An information-value vocabulary

Definitions (after Birge & Louveaux):

$$\text{EVPI} = \mathbb{E}[W^*(\omega)] - W^{\text{RP}} \quad (\text{price of perfect info})$$

$$\text{VSS} = W^{\text{RP}} - W^{\text{EEV}} \quad (\text{value of the SP solution})$$

Plain English.

- EVPI = how much should a planner pay for perfect SAF-price forecasts?
- VSS = how costly is using the mean SAF price as a planning value?

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Headline preview.

$$\text{EVPI (candidate-based estimate)} = \$255 \text{ M/yr} = 19.8\% \text{ of } \mathbb{E}[\text{PVC}].$$

Why this is an *estimate* rather than a certified bound $\rightarrow$ Act IV.

Act III — Closed-Form Benchmark

Why a closed-form?

The 700k-variable MILP is opaque. We collapse to a *2-state* reduction that:

1. Lets a referee replicate the mechanism by hand
2. Provides an analytical anchor for the numerical σ^* in Act IV
3. Delivers a per-unit number directly comparable to 45Z and CORSIA

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In one line. We extend Weitzman (1974) prices-vs-quantities to a *mean-CVaR* planner.

- Weitzman: risk-neutral, symmetric noise $\Rightarrow$ optimal Pigouvian subsidy = 0
- Our 2-state mean-CVaR: $\sigma^* > 0$ proportional to risk aversion λ *and* upside spread

The 2-state setup

Output decision: $Q \in \mathbb{R}_+$ (gallons SAF, year-0 representative).

State: $\omega \in \{H, L\}$, $\mathbb{P}[\omega = H] = \pi_H \in (0, 1)$.

Linear inverse demand: $P^d(Q; \omega) = a(\omega) - bQ$, $b > 0$, $a_H > a_L > c$.

Quadratic capacity cost: $C(Q) = \frac{k}{2}Q^2$.

Net premium: $A(\omega) \equiv a(\omega) - c$, combined curvature $K \equiv b + k > 0$.

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Assumptions (8 total). Two-state demand; linear demand; convex cost; state ordering; positive net premium; α, λ bounded; partial equilibrium.

All assumptions explicit, all σ^* partials *symbolically verified* (17 SymPy assertions PASS).

Lemma 1 — strict concavity & interior optimum

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Under Assumptions 1–7, the mean-CVaR welfare functional

$$\mathcal{W}(Q) = (1 - \lambda) \mathbb{E}[W(Q; \omega)] + \lambda \text{CVaR}_\alpha[W(Q; \omega)]$$

is strictly concave in Q on $\mathbb{R}_+$ with $\mathcal{W}''(Q) = -K < 0$, and the maximizer Q_{risk}^* is interior.

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Sketch.

- Each per-state $W(Q; \omega)$ is strictly concave ($-K$).
- $\mathbb{E}[\cdot]$ is a convex combination $\Rightarrow$ concave.
- $\text{CVaR}_\alpha[\cdot]$ on 2 states is also a convex combination (explicit weights below) $\Rightarrow$ concave.
- Convex combinations of concave functions are concave; positive coefficient on the strict piece $\Rightarrow$ strict

Theorem 1 — piecewise closed-form $\sigma^*(\alpha)$

Theorem 1 (Risk-premium-equivalent subsidy)

Under Assumptions 1–8,

$$\sigma^*(\alpha) = \lambda(a_H - a_L) \min\left\{\frac{\alpha \pi_L}{1 - \alpha}, \pi_H\right\}.$$

In the **collapsed regime** $\alpha \geq \pi_H$:

$$\boxed{\sigma^* = \lambda \pi_H (a_H - a_L),} \quad (\text{independent of } \alpha)$$

In the **split regime** $\alpha < \pi_H$, $\sigma^*(\alpha)$ rises with α .

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Three-factor reading (collapsed regime):

$$\sigma^* = \underbrace{\lambda}_{\text{social risk aversion}} \cdot \underbrace{\pi_H}_{\text{upside prob.}} \cdot \underbrace{(a_H - a_L)}_{\text{demand spread}}.$$

Each factor is observable from the empirical KDE on SAF prices.

Corollary — Madansky EVPI for the 2-state planner

Corollary 1 (Risk-neutral EVPI)

Under Assumptions 1–8 and $\lambda = 0$,

$$\text{EVPI} = \frac{\text{Var}(A)}{2K} = \frac{\pi_H \pi_L (a_H - a_L)^2}{2K}.$$

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Why this matters.

- **Analytical anchor** for SP-supply-chain literature: an EVPI *formula*, not a simulation
- Maxed at $\pi_H = \frac{1}{2}$ (symmetric state mass)
- Sign pattern: $\partial \text{EVPI} / \partial \text{Var}(A) > 0$, $\partial \text{EVPI} / \partial K < 0$

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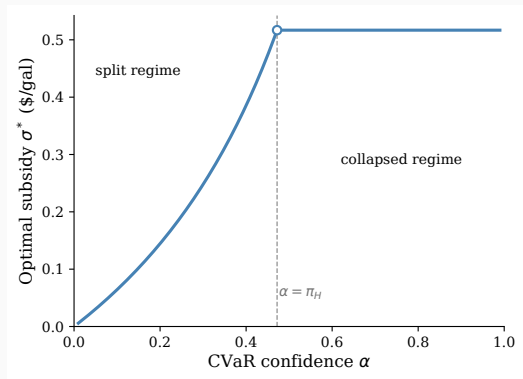
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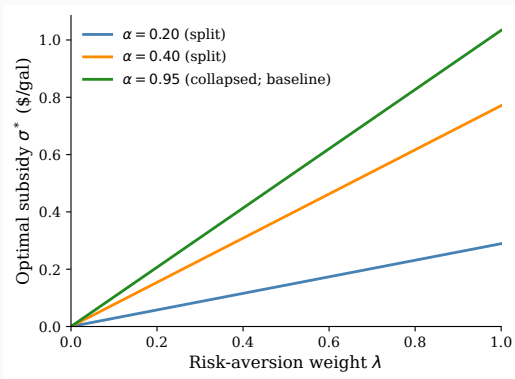
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Provenance. Madansky (1960) for the LP planner; we restore it for a quadratic-cost / linear-demand SP planner under explicit assumptions.

σ^* comparative statics — visual intuition



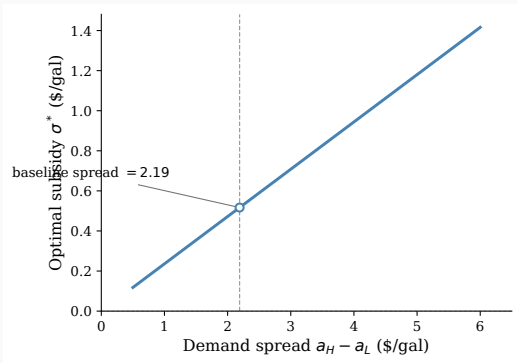
$\sigma^*(\alpha)$ — regime kink at $\alpha = \pi_H$



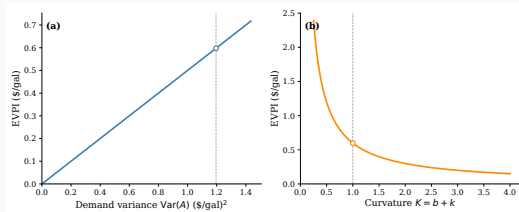
$\sigma^*(\lambda)$ at three α levels

- *Left:* in the collapsed regime, σ^* is α -independent (Cor. 1' continuity).
- *Right:* σ^* is linear in λ , recovering Weitzman ($\lambda = 0 \Rightarrow \sigma^* = 0$).

σ^* comparative statics — demand spread + EVPI



$$\sigma^*(a_H - a_L); \text{ slope } \lambda\pi_H = 0.236$$



EVPI vs. $\text{Var}(A)$ (left), vs. curvature K (right)

Baselines plotted at $\lambda = 0.5$, $\alpha = 0.95$, $\pi_H = 0.472$, $a_H - a_L = 2.19$ \$/gal, $K = 1$.

KDE calibration: from 20 bins to 2 states

Median-split of the 20-bin KDE on SAF spot price at \$6.00/gal:

- $\pi_H = 0.472$ (probability mass above \$6.00/gal)
- $\pi_L = 0.528$
- $a_H = \$7.16/\text{gal}$ (high-state conditional mean)
- $a_L = \$4.97/\text{gal}$ (low-state conditional mean)
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KDE-anchored headline

$$\sigma_{\text{KDE}}^* = \lambda \cdot \pi_H \cdot (a_H - a_L) = 0.5 \times 0.472 \times 2.19 \approx \text{\textcolor{brown}{\$0.52/gal}}.$$

This is the per-unit policy number anchored directly on the empirical SAF-price distribution; the closed-form predicts the toy planner's subsidy, an analytical anchor for the numerical σ^* from the 700k-variable SP.

Act IV — Empirical Stochastic Program

From toy to landscape: the empirical SP-MILP

Recall the planner's discretized problem from Act II. In numbers:

Quantity	Value
Hexagons (50 mi^2 , SE-US frame)	14,040
Counties (FIPS-5, after hex $\rightarrow$ county agg.)	1,374
Depot candidate locations $ \mathcal{D} $	528
Refinery candidate locations $ \mathcal{R} $	528
Stage-2 residue $\rightarrow$ depot pairs ($\leq 50 \text{ mi}$)	875,781
Stage-2 depot $\rightarrow$ refinery pairs ($\leq 75 \text{ mi}$)	46,730
Planning horizon	10 yrs
SAF-price scenarios $ \mathcal{S} $ (KDE bins)	20
Ash-tier categories $A_d \times A_r$	10×10
Continuous variables	$\sim 700,000$
Binary variables	$\sim 10,000$

Aggregation invariance — why we can use counties

Proposition (Aggregation invariance under area-share weighting)

Let area shares $\theta_{h,F}$ split hex h across counties F with $\sum_F \theta_{h,F} = 1$. Define $\text{AvailInv}_F = \sum_h \theta_{h,F} \text{AvailInv}_h$. Setting the county cost coefficient

$$c_F = \kappa \cdot \text{HC} \cdot K_F / (\text{AvailInv}_F + \varepsilon)$$

preserves $\sum_F c_F \bar{x}_F = \sum_h c_h \bar{x}_h$ when $\bar{x}_h \propto \text{AvailInv}_h$.

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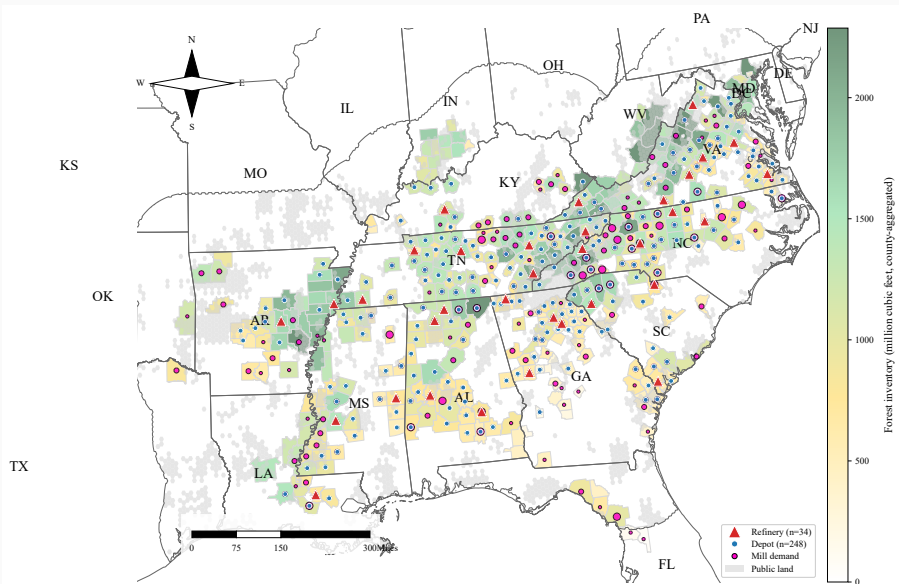
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Why it matters.

- 14,040 hexes $\rightarrow$ 1,374 counties is a *numerical device*, not an economic distortion.
- Empirically verified: county vs. hex total NPV agree within $\pm 0.15\%$.
- Shrinks the DE to $\sim 7 \times 10^5$ continuous + 9.6×10^3 binary variables $\Rightarrow$ Gurobi tractable.

Baseline siting — 248 depots / 34 refineries

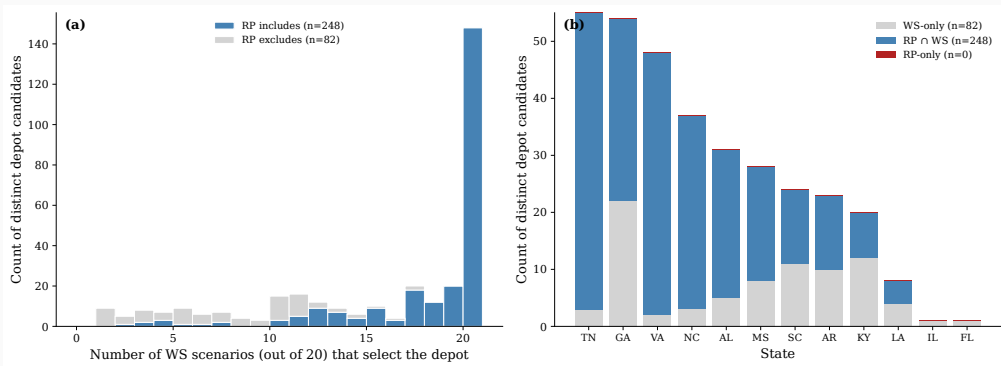


Where the chain lives

State	Depots	%	Refineries	%
Tennessee (TN)	52	21.0%	6	17.6%
Virginia (VA)	46	18.5%	7	20.6%
North Carolina (NC)	34	13.7%	4	11.8%
Georgia (GA)	32	12.9%	4	11.8%
Alabama (AL)	26	10.5%	4	11.8%
Mississippi (MS)	20	8.1%	2	5.9%
South Carolina (SC)	13	5.2%	3	8.8%
Arkansas (AR)	13	5.2%	2	5.9%
Kentucky (KY)	8	3.2%	1	2.9%
Louisiana (LA)	4	1.6%	1	2.9%
Total	248	100%	34	100%

77% of depots and 74% of refineries concentrated in the SE-U.S. core (TN, VA, NC, GA, AL).

RP vs. WS: the recourse policy concentrates the chain



- 100% of the 248 RP depots appear in at least one WS solution.
- WS union: 330 distinct candidate locations; 82 are WS-only (scenario-specific marginal capacity).
- Interpretation: the recourse policy picks *robust* sites and defers scenario-specific capacity to routing flexibility.

EVPI — a candidate-based, bias-disclosed estimate

Quantity	Value (\$M/yr, single-year-equivalent)
$\mathbb{E}[W^*(\omega)]$ wait-and-see expectation	\$1,541 M
W^{RP} recourse-problem optimum (LB)	\$1,286 M
W^{EEV} value of expected-value sol.	n.r. (gap-corrupted)
EVPI, candidate-based estimate (difference of two lower bounds)	\$255 M (19.8%)
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Why an estimate, not a certified bound?

- Per-scenario WS solves attain MIP gap $\leq 5\%$ (Gurobi default).
- In the spirit of Mak–Morton–Wood (1999) / Bayraksan–Morton (2011) candidate evaluation — but without replications, so an *estimate*, not a certified bound (paper App. D).
- Reporting the EVPI *lower bound* or VSS would require comparing two gap-corrupted quantities of opposite sign $\Rightarrow$ statistically unsound (paper App. D).

CVaR — where the chain risk actually lives

Seven candidate plans, evaluated ex post ($\alpha = 0.95$; wait-and-see values certified against the full candidate pool; no mean-CVaR MILP is solved — paper §5.7). The

		central (bin-9) plan	197/27 WS candidate
two frontier plans:	Depots / refineries	248 / 34	197 / 27
	$\mathbb{E}[L]$	\$299 M	\$284 M
	$\text{CVaR}_{0.95}[L]$	\$921 M	\$1,342 M
	$\mathbb{E}[W]$ / $\text{CVaR}_{0.95}[W]$	\$16.99B / \$11.43B	\$17.01B / \$12.34B

CVaR — where the chain risk actually lives

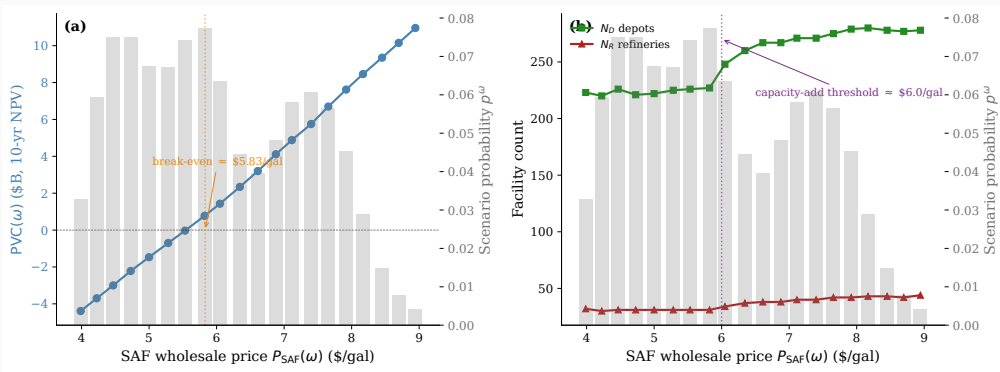
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The reading.

- On *regret*: a genuine two-point frontier — 197/27 wins the mean, 248/34 wins the tail ($\text{CVaR}_{0.95}/\mathbb{E} \approx 3 \Rightarrow$ chain risk is in the tail, not the mean).
- On *welfare levels* (the planner's mean-CVaR object): 197/27 dominates at every $\lambda \in [0, 1]$ — +\$15M/yr expected, +\$0.92B/yr in the 5% tail $\Rightarrow$ the committed (EEV-style) plan overbuilds; contraction is a tail play, as Theorem 1 prices.

Topological response to SAF price



- *Panel (a)*: build-vs-slack break-even at $p_{\text{SAF}} \approx \$5.83/\text{gal}$.
- *Panel (b)*: capacity expansion threshold at $p_{\text{SAF}} \approx \$6.0/\text{gal} \Rightarrow$ depots rise from ~ 225 to ≥ 270 within a narrow band.
- Policy-uncertainty near \$6/gal is the most **capacity-sensitive** neighborhood.

Robustness summary

Things we checked — siting topology and σ^* are stable.

- **Slack-import price.** \$3.50 $\rightarrow$ \$2.85/gal (2025 STEO): $\mathbb{E}[L]$ shifts $< 5\%$, break-even \$5.83 $\rightarrow$ \$5.18/gal, *siting topology unchanged*.
- **OPEXB / elasticity bracket.** 9-cell sweep over $\epsilon \in \{-0.3, -0.5, -1.0\}$ and $\text{OPEXB} \in \{1.96, 2.30, 2.70\}/\text{gal}$: *248/34 siting preserved* across all 9 cells; $\sigma^* \in [\$0.47, \$0.57]/\text{gal}$ ($\pm 10\%$ of \$0.52).
- **Ash-quality uncertainty.** σ perturbation: $\pm \$0.30/\text{gal}$ welfare-equiv. cost at fixed p_{SAF} , an *order of magnitude smaller* than the price-spread effect (\$2.19/gal).
- **Softwood-inclusive pool.** 25.8 $\rightarrow$ 41.6 M dt/yr: *topology preserved*.

Act V — Policy and Conclusion

The 45Z policy bridge — a qualitative comparison

Two numbers in two different equilibria:

- Welfare-frame $\sigma_{\text{KDE}}^* = \$0.52/\text{gal}$ — planner mean-CVaR, $\lambda = 0.5$
- Project-finance 45Z $\approx \$1.30\text{--}\$1.45/\text{gal}$ — GREET 2024 center-case (FT-from-forest-residue, SMR-H₂, mixed-counterfactual)

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Why we do not claim a quantitative bridge.

- Different objects: social planner welfare vs. investor-IRR producer credit
- 45Z carries a *risk-adjusted weighted-cost-of-capital* wedge our representative-planner toy does not estimate
- Both numbers in the same \$0.5–\$2/gal order-of-magnitude band $\Rightarrow$ qualitative consistency, not a calibration

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Plus: GE channels (i)–(iii) push σ_{KDE}^* upward $\Rightarrow$ the planner-side and producer-side

External validity — the formula is path-agnostic

$$\sigma^* = \lambda \pi_H (a_H - a_L)$$

depends only on the high/low-state probabilities and conditional means of the relevant biofuel-price distribution.

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Recipe for other 2G biofuels:

- Corn-stover-to-renewable-diesel (hydroprocessing)
- Soy-oil-to-renewable-diesel (HEFA)
- Cellulosic-gasoline (mixed-alcohol synthesis)

Substitute the appropriate KDE on wholesale price, USDA-ERS / RFS-2 feedstock availability, and the 45Z carbon-intensity tradespace.

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Expected band: ~\$0.20–\$1.50/gal across pathways. The methodology transfers; the numerical anchor is pathway-specific

Limitations & open threads

1. **Single-period investment timing.** 10-year horizon, no deferral. Real-options deferral value bounded above by the value of perfect information ($\approx \$255\text{M/yr}$ candidate-based estimate).
2. **Mill demand exogenous.** Faustmann supply-side response is a coupled Stage-0 problem.
3. **Partial equilibrium.** CGE stumpage / LCFS rents / RIN pass-through held exogenous (channels (i)–(iii)).
4. $p_C = 0$ **here.** Pigouvian carbon tax + 45Z bridge = *companion paper*.
5. **Perfect SAF/fossil substitution.** CES nesting requires substitution-elasticity identification.
6. **Intermediate $\lambda \in (0, 1)$ MILP.** Closed-form predicts linearity; testing it requires longer wall time.

Three follow-up threads

1. **Faustmann + endogenous stumpage.** Couple a forestry Stage-0 to our SAF Stage-1/2 SP $\Rightarrow$ closes channel (i) of the GE adjustment.
2. **45Z + Pigouvian carbon.** Companion paper: $p_C \in \{0, 50, 100, 200\}$ $\$/\text{tCO}_2 \times 45Z \in \{\text{status quo, removed, doubled}\}$, same model architecture.
3. **Continuous-state real options.** A Dixit–Pindyck / Boomsma-style extension that delivers analytical comparative statics for the policy-uncertainty channel our discrete SP discretizes but does not characterize.

Wrap-up — five things to take away

1. A familiar engineering MILP is also a **welfare problem** once you put slack-import opportunity cost back in.
2. Mean-CVaR Weitzman gives a **closed-form** $\sigma^* = \lambda \pi_H(a_H - a_L)$ and a Madansky EVPI $\text{Var}(A)/(2K)$.
3. The empirical SP delivers **248 depots / 34 refineries** across 10 SE-U.S. states; the recourse policy concentrates on robust sites.
4. **EVPI $\approx$ \$255M/yr (19.8%)** reported honestly as a candidate-based estimate with disclosed bias channels.
5. **CVaR_{0.95} / $\mathbb{E} \approx 7$** $\Rightarrow$ chain risk is in the tail, not the mean. Welfare-frame $\sigma^* = \$0.52/\text{gal}$ is the risk component of 45Z's \$1.30–\$1.45/gal (a two-sided bound; paper §6.1).

Thank you.

Backup slides

Available on request: full proof of Theorem 1, KDE bin tabulation, all 9-cell OPEXB sensitivity, full per-state breakdown, App. C MIP-gap argument.

Backup — Proof of Theorem 1 (sketch)

Step 1. FOC of $\mathcal{W}(Q) = A_{\text{eff}}Q - \frac{K}{2}Q^2 \Rightarrow Q_{\text{risk}}^* = A_{\text{eff}}/K$.

Step 2. Compute A_{eff} in two regimes.

- *Collapsed* ($\alpha \geq \pi_H$): CVaR_α weights $\rightarrow (0, 1)$ on L state, $A_{\text{eff}} = (1 - \lambda)\mathbb{E}[A] + \lambda A_L$.
- *Split* ($\alpha < \pi_H$): explicit Rockafellar–Uryasev weights, $A_{\text{eff}} = \mathbb{E}[A] - \frac{\lambda\alpha\pi_L}{1-\alpha}(a_H - a_L)$.

Step 3. Risk-neutral benchmark $Q_{\text{RN}}^* = \mathbb{E}[A]/K$.

Step 4. Subsidy σ shifts the FOC: $Q_{\text{risk}}^*(\sigma) = (A_{\text{eff}} + \sigma)/K$.

Solve $Q_{\text{risk}}^*(\sigma) = Q_{\text{RN}}^* \Rightarrow \sigma^*(\alpha)$ piecewise.

Step 5. Substitution gives

$$\sigma^*(\alpha) = \lambda(a_H - a_L) \min\left\{\frac{\alpha\pi_L}{1-\alpha}, \pi_H\right\}.$$

Backup — KDE bin table (20 SAF-price scenarios)

ω	Left	Right	p^ω	$p_{\text{SAF}}(\omega)$	Cum. p
1	3.81	4.08	0.033	3.99	0.033
2	4.08	4.34	0.059	4.22	0.092
3	4.34	4.61	0.075	4.47	0.167
4	4.61	4.87	0.075	4.73	0.242
5	4.87	5.14	0.068	5.00	0.309
6	5.14	5.40	0.067	5.28	0.377
7	5.40	5.67	0.074	5.53	0.451
8	5.67	5.93	0.077	5.82	0.528
9	5.93	6.20	0.064	6.05	0.591
10	6.20	6.47	0.044	6.34	0.636
11	6.47	6.73	0.040	6.61	0.675
12	6.73	6.99	0.048	6.88	0.724
13	6.99	7.26	0.058	7.11	0.782
14	7.26	7.52	0.061	7.40	0.842
15	7.52	7.79	0.057	7.66	0.899
16	7.79	8.06	0.045	7.92	0.944
17	8.06	8.32	0.029	8.17	0.973
18	8.32	8.59	0.015	8.45	0.988
19	8.59	8.85	0.008	8.70	0.996
20	8.85	9.12	0.004	8.94	1.000
<i>Mean (prob-weighted)</i>			1.000	6.00	—

EIA STEO 2014–2024 wholesale Gulf-Coast jet-A, Silverman bandwidth.

Backup — what the \$255M is, and is not (paper App. D)

What is actually computed pairs two lower bounds.

$$\mathbb{E}[W^*]_{\text{LB}} = \sum_{\omega} p^{\omega} \text{LB}_{\omega} \quad (\text{incumbent-weighted; } \leq \mathbb{E}[W^*])$$

$$\max_c \mathbb{E}[W \mid y_c] \leq W^{\text{RP}^*} \quad (\text{best candidate; extensive form not solved})$$

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Honest reading.

- A certified one-sided bound would need the best-bound-weighted $\sum_{\omega} p^{\omega} \text{UB}_{\omega}$ — not delivered at the low-price bins (median realized gap 8.07%).
- The difference of two lower bounds has **no determinate bias sign**.
- $\Rightarrow$ **\$255M is a candidate-based EVPI estimate**, not a certified bound.

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Why VSS is n.r. W^{EEV} requires a *point estimate* comparison with W^{RP} , both of opposite-direction gap bias $\Rightarrow$ statistically unsound under our wall-time budget.

Backup — References (selected)

- **Welfare/policy:** Weitzman (1974); Holland et al. (2009); Lade & Bushnell (2019); Hertel et al. (2010); Searle (2024); Lohawala, Toman & Joiner (2024); Jiang & Liu (2025).
- **Risk:** Rockafellar & Uryasev (2000, 2002); Artzner et al. (1999); Conejo et al. (2010); Boomsma et al. (2012); Insley (2013).
- **Stochastic programming:** Birge & Louveaux (1997); Shapiro et al. (2014); Madansky (1960); Mak, Morton & Wood (1999); Bayraksan & Morton (2011); Santos et al. (2019).
- **Biofuel SP-MILPs:** Eksioglu et al. (2009); Marvin et al. (2012); Lin et al. (2014); Wang et al. (2017); Osmani & Zhang (2017); Quddus et al. (2018); Gebreslassie et al. (2012); Masum et al. (2023).
- **SAF/forest:** NREL Davis et al. (2018); Tan et al. (2017); Argonne GREET (2024); ICAO CORSIA (2019); Langholtz et al. (2024) BT23; USDA (2023); Butler et al. (2016); Faustmann (1849); Amacher et al. (2009); Li (2025).